Appendix 1
Example for treatment of heel angles and water heads at different stages of

---CHUNK---

SOURCE: MSC.245(83).pdf
4 Alternatives 5

Appendix 1
Example for treatment of heel angles and water heads at different stages of
cross-flooding 6

Appendix 2
Friction coefficients in cross-flooding arrangement 7

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Example using figures for a passenger ship 10

RESOLUTION MSC.245(83)
(adopted on 12 October 2007)
RECOMMENDATION ON A STANDARD METHOD FOR
EVALUATING CROSS-FLOODING ARRANGEMENTSMSC 83/28/Add.2
ANNEX 16

I:\MSC\83\28-Add-2.doc
1 Definitions

k : Sum of friction coefficients in the considered cross-flooding arrangement.

S ( )2m : Cross-section area of the cr oss-flooding pipe or duct. If the cross-section area is not circular, then:

4
2
equiv
equiv
D
S

=

where:
p
ADequiv = 4

A = actual cross-section area

p = actual cross-section perimeter

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0 ()° : Angle before commencement of cros s-flooding. This assumes that the cross-flooding device is fully flooded but that no water has entered into the equalizing compartment on the opposite side of the damage (see appendix 1).

f ()° : Heel angle at final equilibrium (f).

()° : Any angle of heel between the commen cement of cross-flooding and the final equilibrium at a given time.

fW ( )3m : Volume of water which is used to bring the ship from commencement of cross-flooding 0 to final equilibrium f .

W ( )3m : Volume of water which is used to br ing the ship from any angle of heel to the final equilibrium f .

0H ()m : Head of water before commencement of cross-flooding, with the same assumption as for 0 .

H ()m : Head of water when any angle of heel is achieved.

fh ()m : Final head of water after cross-flooding (0=fh , when the level inside the

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SOURCE: MSC.245(83).pdf
as for 0 .

H ()m : Head of water when any angle of heel is achieved.

fh ()m : Final head of water after cross-flooding (0=fh , when the level inside the

equalizing compartment is equal to the free level of the sea).

2 Formulae

2.1 Time required from comm encement of cross-flooding 0 to the final equilibrium f :

=
0
0
0
1
1
2
1
2
H
hgH
H
h
F S
W
T
f
f
f
f

2.2 Time required to bring the ship from any angle of heel to the final equilibrium f :

=

H
hgH
H
h
F S
WT
f
f
1
1
2
1
2